DAY - 4 Documentation

🟢 IoT — Thursday
TASK
IoT-Day 4: IMU Data Simulation & Processing
Since you don't have physical sensors and the exact IMU may not be available in Wokwi, we won't depend on a real IMU component today.
Instead, we'll learn how IMU data looks and how the ESP32 firmware can process it.
Don't worry if the simulator doesn't have the exact sensor. You can use simulated/test values for today's task.
Step 1 — Understand the IMU data
Briefly understand:

- Accelerometer
- Gyroscope
- X, Y, Z axes
- Acceleration (g)
- Angular velocity (°/s)

You don't need to study deeply. Just understand what each value represents.
Step 2 — Create sample IMU data
Since there is no physical IMU, create a few test IMU readings in your ESP32 program.
For example:
AX = 0.02 g
AY = 0.01 g
AZ = 0.98 g
GX = 0.5 °/s
GY = 0.2 °/s
GZ = 1.0 °/s
These are TEST/SIMULATED values, not real sensor readings.
Try creating different values for:
A. Stationary
AX ≈ 0
AY ≈ 0
AZ ≈ 1
B. Tilt / rotation
Change some X/Y/Z and gyroscope values.
C. Movement
Create larger changes in the values.
You don't need to make these values physically perfect. The purpose is to understand how changing sensor values would look in the firmware.
Step 3 — Print the data
Use ESP32 + Arduino IDE and print:
AX:
AY:
AZ:
GX:
GY:
GZ:

Update the values at a fixed interval, for example every 500 ms or 1 second.
Your Serial Monitor should continuously show the changing test data.
Step 4 — Calculate acceleration magnitude
Use:
Acceleration Magnitude = $\sqrt{(AX^2 + AY^2 + AZ^2)}$
Display the calculated value along with the IMU values.
Example:
AX: 0.02
AY: 0.01
AZ: 0.98
GX: 0.5
GY: 0.2
GZ: 1.0
Acceleration Magnitude: 0.98 g
Step 5 — Observe the changes
refer the image given

IMU : IMU stands for Inertial Measurement Unit.

It is an electronic sensor system that measures how an object accelerates and rotates.

Imagine attaching an IMU to a car. As the car moves, turns, brakes, or tilts, the IMU produces measurements that help the ESP32 understand the vehicle's motion.

For project :

- GPS tells you where the vehicle is.
- IMU tells you how the vehicle is moving.
- ESP32 reads and processes both types of data.

A typical six-axis IMU contains two sensors.

1. Accelerometer

Measures acceleration along three axes: X, Y and Z.

Unit: g or m/s^2

Example uses: **detecting sudden braking, impact, vibration and tilt.**

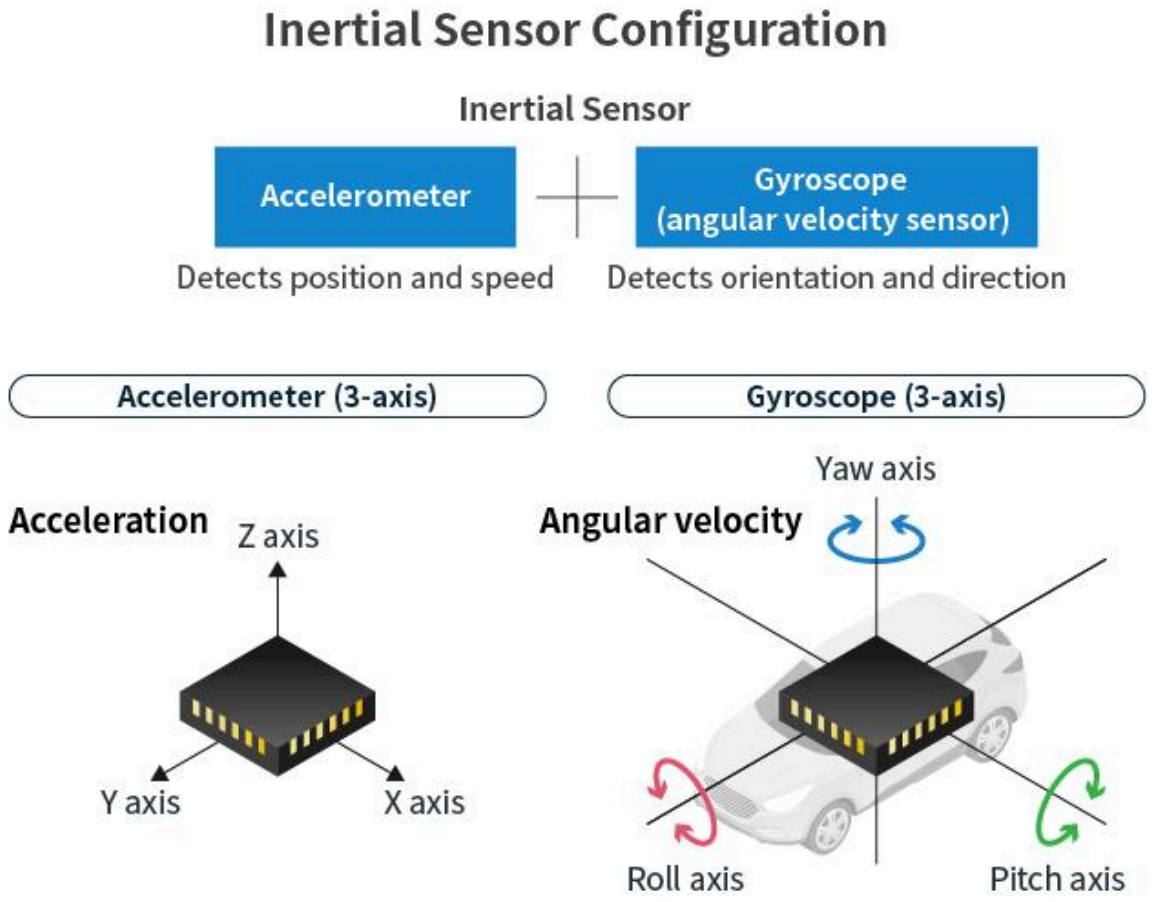
2. Gyroscope

Measures angular velocity around the X, Y and Z axes.

Unit: degrees per second (°/s) or radians per second

Example uses: **detecting turning, rotation and rapid changes in orientation.**

Some IMUs also include a magnetometer, which measures the magnetic field and can help estimate heading. An accelerometer plus gyroscope is commonly called a six-axis IMU; adding a three-axis magnetometer gives a nine-axis sensor system.



X-axis: forward and backward.
Y-axis: left and right.
Z-axis: up and down.

- Accelerometer = linear acceleration : Measures acceleration and responds to gravity. Useful for impact, braking, vibration and tilt estimation.
- Gyroscope = angular velocity: Measures rotational speed. Useful for turns, rapid rotation and orientation tracking.

Accelerometer example

Suppose the vehicle is stationary on a flat surface, and the sensor's Z-axis points upward.

Axis	Example reading	Interpretation
AX	0.02 g	Almost no acceleration along X
AY	0.01 g	Almost no acceleration along Y
AZ	0.98 g	Approximately one g of gravity measured along Z

Why is AZ close to 1 g even though the vehicle is stationary? Because an accelerometer measures specific force, which includes the effect of gravity. A stationary sensor commonly reads a magnitude near 1 g.

Gyroscope reading	Meaning
GX = 0°/s	No rotation detected around X
GY = 10°/s	Rotating around Y at 10 degrees per second
GZ = 90°/s	Rotating around Z at 90 degrees per second

If a sensor rotates steadily at 90°/s around one axis for one second, the rotation would be about 90 degrees under that simplified condition.

In real systems, integrating gyro readings to estimate orientation can accumulate errors called drift.

Situation	What the IMU may detect	What GPS contributes
Sudden braking	Strong acceleration change	Speed and position
Impact or collision	A sudden acceleration spike	Vehicle location
Sharp turn	Rotation and lateral acceleration	Position and course
Vehicle tilting or rollover	Changing orientation and rotation	Location, when available

1. Generate six test values: AX, AY, AZ, GX, GY and GZ.
2. Create stationary, tilt/rotation and movement scenarios.
3. Print all six readings in the Serial Monitor.
4. Calculate acceleration magnitude:

$$a = \sqrt{AX^2 + AY^2 + AZ^2}$$

5. Clearly document that the readings are SIMULATED / TEST DATA, not real sensor measurements.

Test A

Stationary

TEST CASE: STATIONARY
Data: SIMULATED / TEST DATA
AX: 0.02 g

AY: 0.01 g
AZ: 0.98 g
GX: 0.50 deg/s
GY: 0.20 deg/s
GZ: 1.00 deg/s
Acceleration Magnitude: 0.980 g
Test B

Tilt / Rotation

TEST CASE: TILT / ROTATION
AX: 0.35 g
AY: 0.15 g
AZ: 0.92 g
GX: 12.00 deg/s
GY: 5.00 deg/s
GZ: 3.00 deg/s
Acceleration Magnitude: 0.999 g
Test C

Movement

TEST CASE: MOVEMENT
AX: 1.20 g
AY: 0.40 g
AZ: 0.80 g
GX: 45.00 deg/s
GY: 20.00 deg/s
GZ: 10.00 deg/s
Acceleration Magnitude

Understand and verify the calculation

The acceleration magnitude combines the X, Y and Z readings:

$$a = \sqrt{AX^2 + AY^2 + AZ^2}$$

For the movement test:

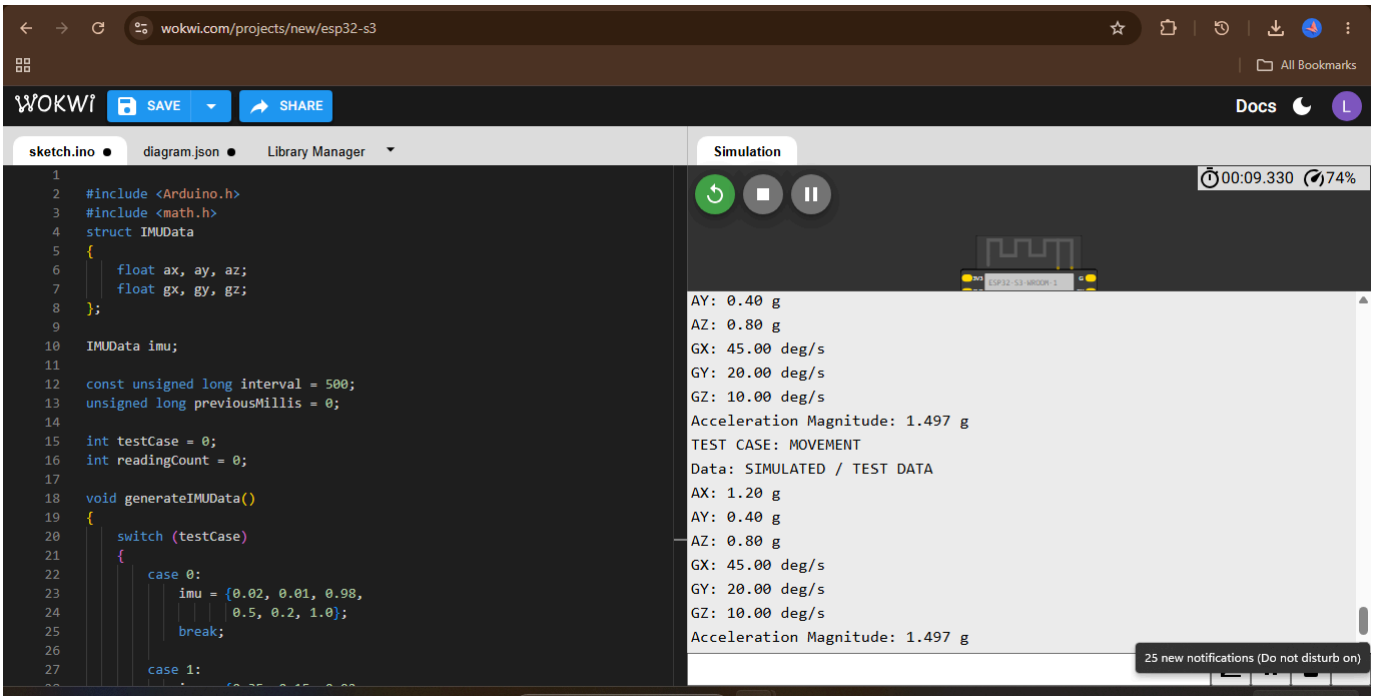
$$a = \sqrt{1.20^2 + 0.40^2 + 0.80^2}$$

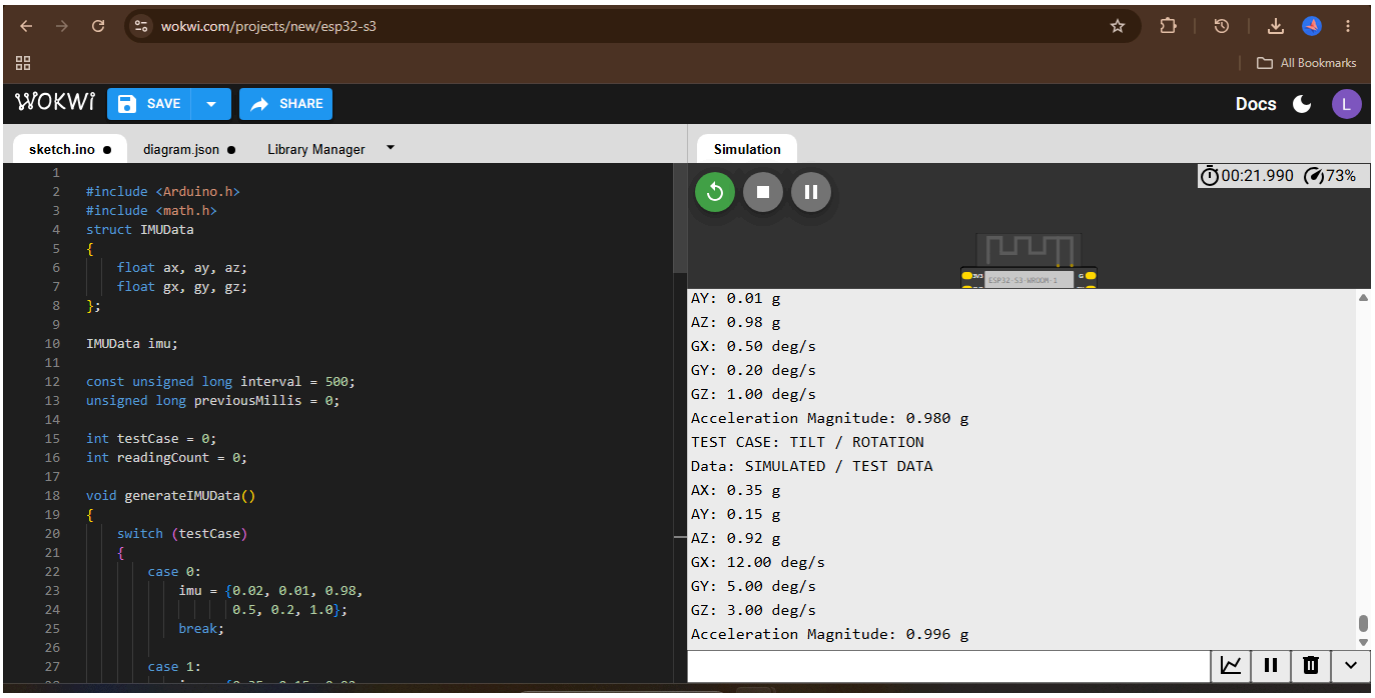
$$a = \sqrt{1.44 + 0.16 + 0.64} = \sqrt{2.24} \approx 1.497g$$

Notice that the magnitude is one combined value. It doesn't tell you which direction the vehicle is moving.

Step 5 — Record your observations

Test case	What you should observe
Stationary	Acceleration magnitude is close to 1 g; gyro values are small.
Tilt / rotation	Acceleration is distributed across the axes; gyro values are higher.
Movement	The selected test has larger acceleration and angular velocity values.




wokwi.com/projects/new/esp32-s3
All Bookmarks
WOKWI
SAVE
SHARE
Docs
sketch.ino
diagram.json
Library Manager
Simulation
00:21.990
73%
#include <Arduino.h>
#include <math.h>
struct IMUData
{
float ax, ay, az;
float gx, gy, gz;
};
IMUData imu;
const unsigned long interval = 500;
unsigned long previousMillis = 0;
int testCase = 0;
int readingCount = 0;
void generateIMUData()
{
switch (testCase)
{
case 0:
imu = {0.02, 0.01, 0.98,
0.5, 0.2, 1.0};
break;
case 1:
AY: 0.01 g
AZ: 0.98 g
GX: 0.50 deg/s
GY: 0.20 deg/s
GZ: 1.00 deg/s
Acceleration Magnitude: 0.980 g
TEST CASE: TILT / ROTATION
Data: SIMULATED / TEST DATA
AX: 0.35 g
AY: 0.15 g
AZ: 0.92 g
GX: 12.00 deg/s
GY: 5.00 deg/s
GZ: 3.00 deg/s
Acceleration Magnitude: 0.996 g